WORKED SOLUTION

STEP II 2010, Question 11

Mechanics • Three-Force Equilibrium

STEP II 2010, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q11

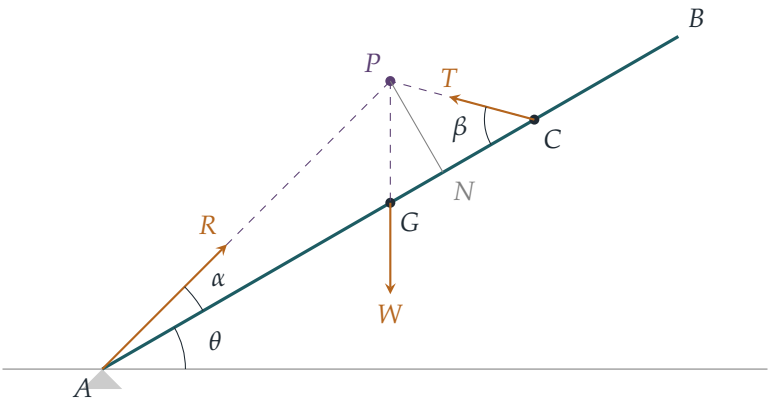
The key idea

Only three forces act on the rod: its weight, the reaction at A and the tension at C. When three non-parallel forces keep a body in equilibrium, their lines of action must pass through a single point. So the line of the reaction from A, the line of the string from C and the vertical through the midpoint of the rod all meet. Measuring distances along and perpendicular to the rod from that common point turns the whole question into a little right-angled trigonometry, with no need to find the forces themselves.

Solution

The model and the diagram

We model the rod as a uniform rigid rod, so its weight W acts at its midpoint G, with $AG = 2L$. The string is light, so it pulls on the rod at C along its own length, where $AC = 3L$. The support exerts a single force R at A (we are not told whether it is smooth or rough, and we don't need to be). The diagram shows the case $\theta = 30°$, $\beta = 45°$ to scale.



Showing the formula

The weight acts vertically through G; the string's line of action passes through C and the reaction's through A. These three lines are not parallel, so all three pass through one point, which we call P.

Ask yourself Why must the three lines of action meet at a point?

Let the lines of action of R and T meet at P. Taking moments about P, both R and T have zero moment, so for the rod to be in rotational equilibrium the weight must also have zero moment about P. That means the vertical line through G passes through P too.

Let N be the foot of the perpendicular from P to the rod, and let $h = PN$. In the right-angled triangle PNA the angle at A is α, so $AN = h\cot\alpha$. In the right-angled triangle PNC the angle at C is β, so $NC = h\cot\beta$. Since $AN + NC = AC$,

$$h\cot\alpha + h\cot\beta = 3L. \tag{1}$$

(If α or β is obtuse, N lies beyond A or C, but the signed lengths $h\cot\alpha$ and $h\cot\beta$ still add up to AC, so (1) holds in every case.)

Now use the third line of action. The rod makes angle θ with the horizontal, so the perpendicular PN makes angle θ with the vertical PG. In the right-angled triangle PNG the angle at P is θ, so $GN = h\tan\theta$, and N lies on the upper side of G (further from A), as in the diagram. Hence

$$h\cot\alpha = AN = AG + GN = 2L + h\tan\theta. \tag{2}$$

Subtracting (2) from (1) gives $h\cot\beta = L - h\tan\theta$, that is,

$$L = h(\cot\beta + \tan\theta).$$

Substituting this value of L into (2):

$$h\cot\alpha = 2h(\cot\beta + \tan\theta) + h\tan\theta = h(3\tan\theta + 2\cot\beta).$$

Since $h \neq 0$ (otherwise P would lie on the rod at G, and the string, which is attached at C, would lie along the rod, so it could not balance the moment of the weight about A), we may divide by h:

$$\cot\alpha = 3\tan\theta + 2\cot\beta,$$

as required.

The case $\theta = 30°$, $\beta = 45°$

With $\tan 30° = \frac{1}{\sqrt{3}}$ and $\cot 45° = 1$,

$$\cot\alpha = \frac{3}{\sqrt{3}} + 2 = 2 + \sqrt{3}.$$

Now we need $\cot 15^\circ$. Using $\tan(A-B)$ with $A = 45^\circ$, $B = 30^\circ$,

$$\tan 15^\circ = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{2} = 2 - \sqrt{3},$$

so

$$\cot 15^\circ = \frac{1}{2-\sqrt{3}} = \frac{2+\sqrt{3}}{(2-\sqrt{3})(2+\sqrt{3})} = 2 + \sqrt{3}.$$

Hence $\cot\alpha = \cot 15^\circ$. The angle α lies between 0° and 180°, and cot is strictly decreasing on that interval, so it takes each value exactly once. Therefore $\alpha = 15^\circ$.

Checks

- *An independent route.* Taking moments about A gives $T \cdot 3L\sin\beta = W \cdot 2L\cos\theta$. Resolving along and perpendicular to the rod then gives the components of R, and $\cot\alpha$ is their ratio; this reproduces the formula. We checked it numerically this way for 10^5 random pairs (θ, β), with agreement to about 10^{-15}.
- *Special values.* For $\theta = 30^\circ$, $\beta = 45^\circ$ the numerical equilibrium gives $\alpha = 15.000^\circ$, and $\cot 15^\circ = 3.7320\ldots = 2 + \sqrt{3}$. The diagram is drawn to scale: P lies directly above G, at height $\sqrt{3}L$ above A.
- *Limiting case.* If $\theta \to 0$ and $\beta = 90^\circ$ (horizontal rod, vertical string), the formula gives $\cot\alpha = 0$, so $\alpha = 90^\circ$: the reaction is vertical too, as it must be when every other force is vertical.

Common slips

- Putting the weight at C, or taking $AG = 3L/2$: the weight acts at the midpoint of the whole rod, $2L$ from A.
- Getting the sign of GN wrong. Because the rod slopes upwards from A, the vertical through G meets the perpendicular from P on the side of G away from A, so $AN = 2L + h\tan\theta$, not $2L - h\tan\theta$.
- Stopping at $\cot\alpha = 2 + \sqrt{3}$ without showing that this is $\cot 15^\circ$, and without saying why it fixes α uniquely.

Beyond STEP

The three-force theorem is a special case of a general fact in statics. Any system of forces on a rigid body reduces to a single force through a chosen point plus a couple, and equilibrium means both are zero. At university this becomes the vector condition that the total force and the total moment $\sum \mathbf{r}_i \times \mathbf{F}_i$ both vanish, which then underpins rigid-body dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops torque and equilibrium in this vector language.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.